Control Architecture

↳ The distribution pump (cycling hot water) will run at rated speed to get 8m/s and overcome head.

• Duty cycle operation.

Why? ⟶ To keep the logic simple and avoid cost escalation with PID controllers (cannot use PLC, it is distributed architecture), the system will operate on a temperature band.

For H_A,

Palm Oil is regulated at 40° as per calculations. solidifies around 35C

∴ ± 3°C will be fine.

Simply, IF TEMP H_A < 37°C, divert water there untill 43°C.

In case bulk cooling.

P1 : H_A maintainance
P2 : bulk ~~cool~~ heating

⟶ continiously heat P2 but when P1 (H_A needed), priortise P1 untill done.

REFINEMENT

when P1 active, no need to supply at v= 8ms⁻¹. supply at steady 2ms⁻¹. From my prev. calculations, maintaining a 5KL tank, insulated needs aroun 2 KW max.

I will program the PLC ~~to run~~ / VFD to run to meet around 2K.W. power delivery with 2ms⁻¹ speed. Duty cycle like PWM.

Now, designing for the smaller H_A

By NTU, we need around 2 KW energy delivered, we can use a small coil, but for emergency use, we should keep a larger coil.

condition: diverting flow.

100%

$\longrightarrow$ $3\,ms^{-1}$ $\qquad\qquad$ $\longrightarrow$ $1.28\,ms^{-1}$

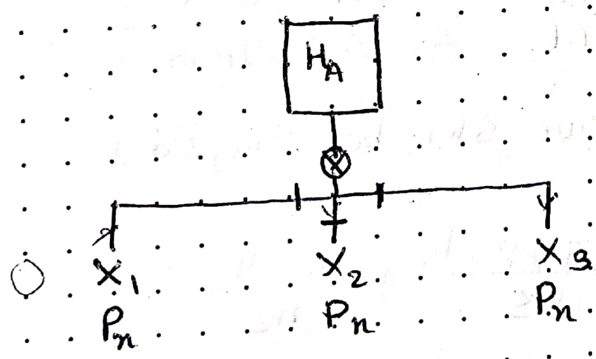
$A = \frac{\pi d^2}{4}$ $\qquad\qquad$ $A = 1\frac{1}{2}''$ pipe.

$= 6.09 \times 10^{-4}$

$= 1''$ pipe.

We cannot distribute the power effectively. So we will operate on discrete operations.

Distribution of palm oil.



← 3 vaulves operated by control

control? → HMI

Let

$f = \alpha$ (flow)

N = need of ports

$\therefore \alpha = N_1 + N_2 + N_3$

N_1

N_2 $\qquad \leftarrow \alpha$

N_3

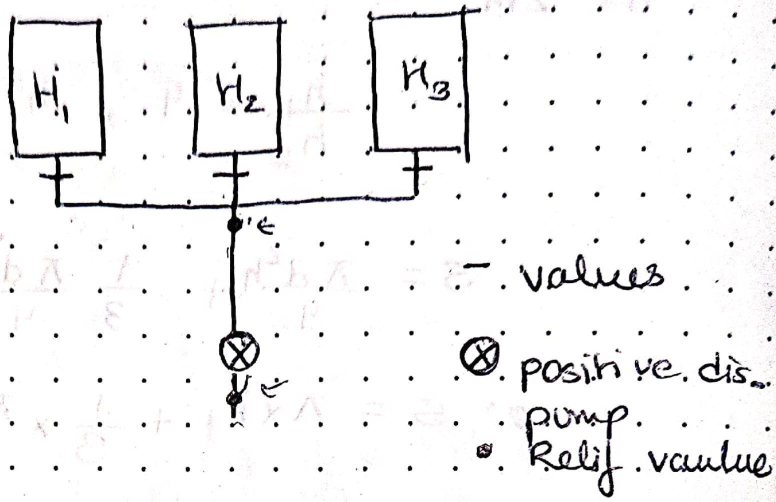
If $N_1 + N_2 + N_3 > \alpha$ use priority basis.

$P_1 > P_2 > P_3$

Distribution/collection of Palm oil

P I) Collection and retention loaded in (H_1, H_2, H_3)

II) The ~~m.~~ oil is melted and transported to H_A



↳ ~~Pump~~. Tank selection - based on level reading.

Retention Tank : $d = 1.8\,m$
$h = 2\,m$